

5bez6zdpv

September 2, 2024

```
[1]: ! pip install transformers
```

```
Requirement already satisfied: transformers in /opt/conda/lib/python3.10/site-  
packages (4.44.0)  
Requirement already satisfied: filelock in /opt/conda/lib/python3.10/site-  
packages (from transformers) (3.15.1)  
Requirement already satisfied: huggingface-hub<1.0,>=0.23.2 in  
/opt/conda/lib/python3.10/site-packages (from transformers) (0.24.6)  
Requirement already satisfied: numpy>=1.17 in /opt/conda/lib/python3.10/site-  
packages (from transformers) (1.26.4)  
Requirement already satisfied: packaging>=20.0 in  
/opt/conda/lib/python3.10/site-packages (from transformers) (21.3)  
Requirement already satisfied: pyyaml>=5.1 in /opt/conda/lib/python3.10/site-  
packages (from transformers) (6.0.2)  
Requirement already satisfied: regex!=2019.12.17 in  
/opt/conda/lib/python3.10/site-packages (from transformers) (2024.5.15)  
Requirement already satisfied: requests in /opt/conda/lib/python3.10/site-  
packages (from transformers) (2.32.3)  
Requirement already satisfied: safetensors>=0.4.1 in  
/opt/conda/lib/python3.10/site-packages (from transformers) (0.4.4)  
Requirement already satisfied: tokenizers<0.20,>=0.19 in  
/opt/conda/lib/python3.10/site-packages (from transformers) (0.19.1)  
Requirement already satisfied: tqdm>=4.27 in /opt/conda/lib/python3.10/site-  
packages (from transformers) (4.66.4)  
Requirement already satisfied: fsspec>=2023.5.0 in  
/opt/conda/lib/python3.10/site-packages (from huggingface-  
hub<1.0,>=0.23.2->transformers) (2024.6.1)  
Requirement already satisfied: typing-extensions>=3.7.4.3 in  
/opt/conda/lib/python3.10/site-packages (from huggingface-  
hub<1.0,>=0.23.2->transformers) (4.12.2)  
Requirement already satisfied: pyparsing!=3.0.5,>=2.0.2 in  
/opt/conda/lib/python3.10/site-packages (from packaging>=20.0->transformers)  
(3.1.2)  
Requirement already satisfied: charset-normalizer<4,>=2 in  
/opt/conda/lib/python3.10/site-packages (from requests->transformers) (3.3.2)  
Requirement already satisfied: idna<4,>=2.5 in /opt/conda/lib/python3.10/site-  
packages (from requests->transformers) (3.7)  
Requirement already satisfied: urllib3<3,>=1.21.1 in
```

```
/opt/conda/lib/python3.10/site-packages (from requests->transformers) (1.26.18)
Requirement already satisfied: certifi>=2017.4.17 in
/opt/conda/lib/python3.10/site-packages (from requests->transformers) (2024.7.4)
```

```
[2]: from tensorflow.keras.models import Model, Sequential
      from tensorflow.keras.layers import Dense, Dropout, Flatten, Conv2D,
      ↪MaxPooling2D, Input
      from tensorflow.keras.callbacks import EarlyStopping
      from tensorflow.python.ops.numpy_ops import np_utils
      from transformers import BertModel, TFBertModel
      import tensorflow as tf
      from tensorflow.keras.optimizers import Adam
      from transformers import BertTokenizer, TFBertModel
      from tensorflow.keras import regularizers
      from sklearn.metrics import classification_report
      from sklearn.metrics import confusion_matrix
```

```
[3]: import pandas as pd
      from matplotlib import rcParams
      import seaborn as sns
      import numpy as np
      from PIL import Image
      from wordcloud import WordCloud
      from sklearn.model_selection import train_test_split
      from matplotlib import pyplot as plt
      from transformers import AutoTokenizer
```

1 Load dataset

```
[4]: fake_data=pd.read_csv("/kaggle/input/fake-news-detection-datasets/News _dataset/
      ↪Fake.csv")
      real_data=pd.read_csv("/kaggle/input/fake-news-detection-datasets/News _dataset/
      ↪True.csv")
```

2 Data Analysis

```
[5]: fake_data.head()
```

```
[5]:
```

	text	label	title \
0	Donald Trump Sends Out Embarrassing New Year'...	0	
1	Drunk Bragging Trump Staffer Started Russian ...	0	
2	Sheriff David Clarke Becomes An Internet Joke...	0	
3	Trump Is So Obsessed He Even Has Obama's Name...	0	
4	Pope Francis Just Called Out Donald Trump Dur...	0	

		text	subject	\
0	Donald Trump just couldn't wish all Americans ...		News	
1	House Intelligence Committee Chairman Devin Nu...		News	
2	On Friday, it was revealed that former Milwauk...		News	
3	On Christmas day, Donald Trump announced that ...		News	
4	Pope Francis used his annual Christmas Day mes...		News	

	date
0	December 31, 2017
1	December 31, 2017
2	December 30, 2017
3	December 29, 2017
4	December 25, 2017

```
[6]: real_data.head()
```

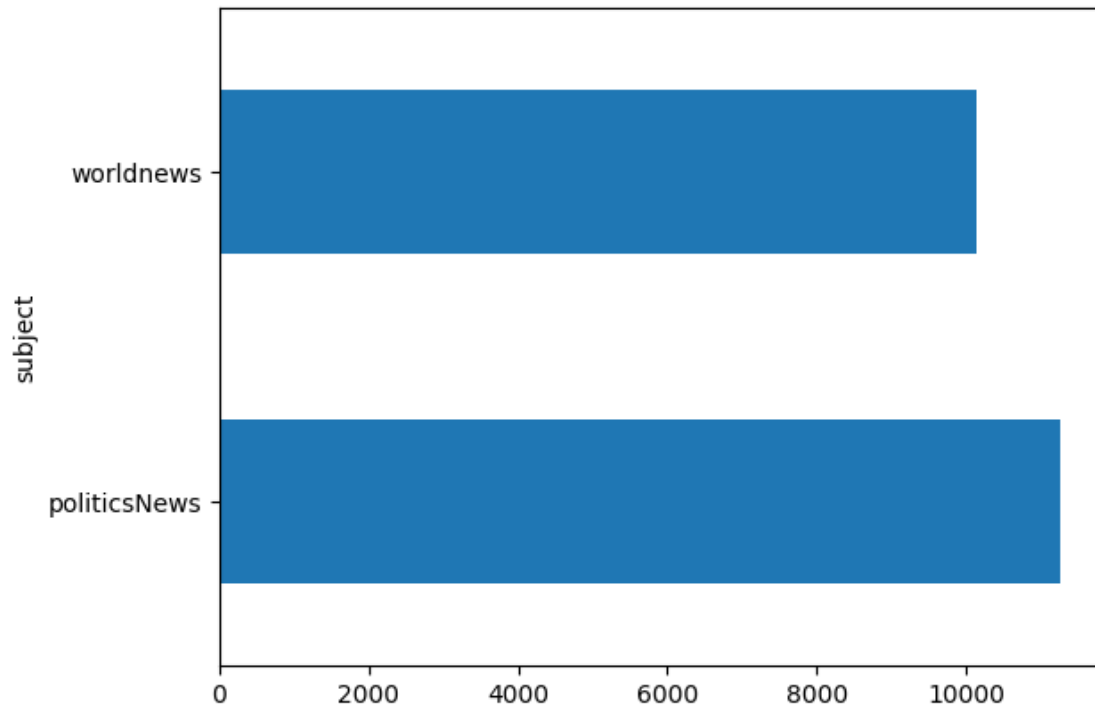
```
[6]:
```

		title	\
0	As U.S. budget fight looms, Republicans flip t...		
1	U.S. military to accept transgender recruits o...		
2	Senior U.S. Republican senator: 'Let Mr. Muell...		
3	FBI Russia probe helped by Australian diplomat...		
4	Trump wants Postal Service to charge 'much mor...		

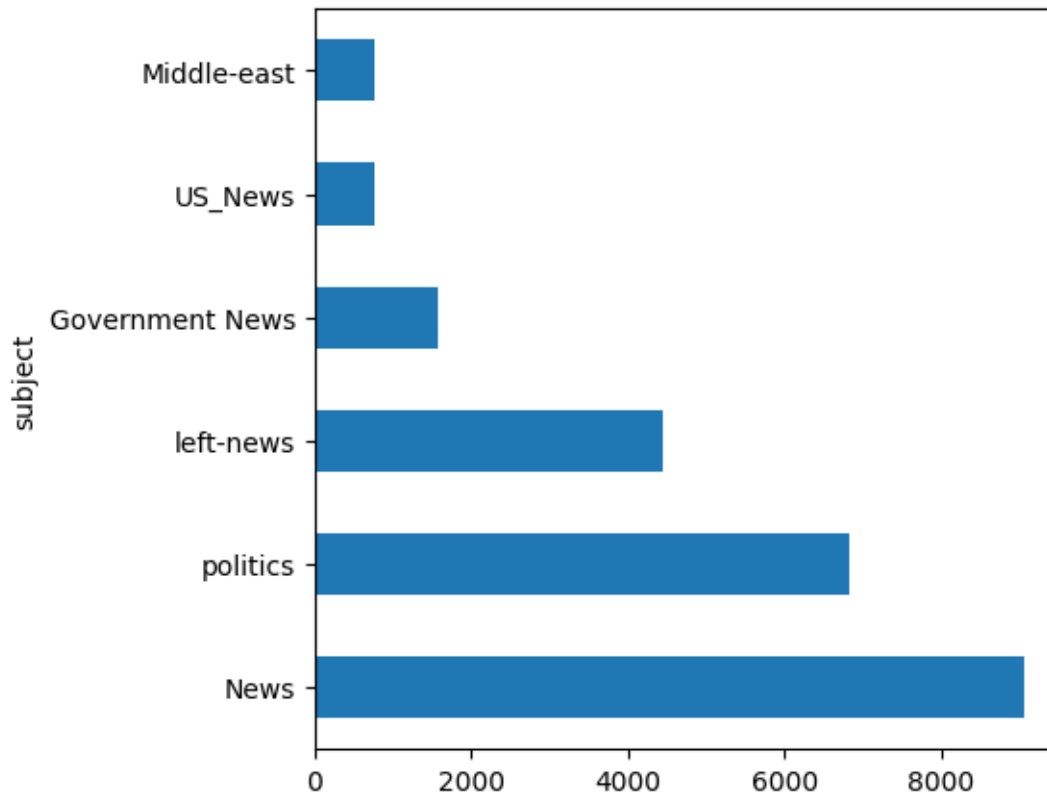
		text	subject	\
0	WASHINGTON (Reuters) - The head of a conservat...		politicsNews	
1	WASHINGTON (Reuters) - Transgender people will...		politicsNews	
2	WASHINGTON (Reuters) - The special counsel inv...		politicsNews	
3	WASHINGTON (Reuters) - Trump campaign adviser ...		politicsNews	
4	SEATTLE/WASHINGTON (Reuters) - President Donal...		politicsNews	

	date
0	December 31, 2017
1	December 29, 2017
2	December 31, 2017
3	December 30, 2017
4	December 29, 2017

```
[7]: real_data['subject'].value_counts().plot(kind='barh')
rcParams['figure.figsize'] = 5,5
```



```
[8]: fake_data['subject'].value_counts().plot(kind='barh')
     rcParams['figure.figsize'] = 5,5
```



3 wordcloud real text data

```
[9]: wordcld_mask=np.array(Image.open("/kaggle/input/wordcloud-mask-collection/
↳twitter.png"))
```

```
[10]: wordcld_text=" ".join(real_data["text"].tolist())
```

```
[11]: wordcloud=WordCloud(width=1920,height=1080,mask=wordcld_mask).
↳generate(wordcld_text)
fig=plt.figure(figsize=(8,8))
plt.imshow(wordcloud)
plt.axis("off")
plt.tight_layout(pad=0)
plt.show()
```



```

8 BREAKING: Trump Tax Documents Leaked And They...
9 Goldman-run funds most popular with Republican...

```

```

                                text      subject \
0 OUR PREVIOUS REPORTS ON JEHMU GREENE:I will NE...      politics
1 WASHINGTON (Reuters) - Congressional Democrats... politicsNews
2 At this point, the arguments against the #AllL...      News
3 MOSCOW (Reuters) - Russia s Foreign Ministry s...      worldnews
4 One of the largest environmental disasters in ...      News
5 MADRID (Reuters) - The Spanish government said...      worldnews
6 ISTANBUL (Reuters) - A Turkish professor who h...      worldnews
7 One of the world s longest-serving and most ic...      Middle-east
8 The chorus of voices saying, Show us your tax...      News
9 NEW YORK (Reuters) - Presidential hopeful Ted ...      politicsNews

```

```

                                date label
0          Jan 16, 2017  fake
1    February 14, 2017  real
2          May 21, 2016  fake
3    September 17, 2017  real
4          January 4, 2016  fake
5    September 15, 2017  real
6          December 1, 2017  real
7    November 26, 2016  fake
8          October 1, 2016  fake
9    January 21, 2016  real

```

```
[18]: final_data.isnull().sum()
```

```

[18]: title      0
      text      0
      subject   0
      date      0
      label     0
      dtype: int64

```

```
[19]: final_data.drop(["subject","date"], axis=1)
```

```

[19]:                                title \
0      FIRE THIS FOX NEWS ANALYST: Trump Was "Alleged...
1      House Democrats escalate effort to obtain Trum...
2      This Tweet Perfectly Connects #AllLivesMatter...
3      Russia's Lavrov and Tillerson talk Syria after...
4      The Worst Environmental Disaster Since The BP...
...
44893  France's Sapin says Brexit would affect London...
44894  LEAKED TAPE Exposes George Soros, Obama GROUP ...

```



```

44895 Trump says current Fed chair Yellen is 'excell...
44896 'Everything but Hillary Clinton,' France's Le ...
44897 Conservative 'Christian' Preacher Wants Gover...

```

```

                                text label
0      OUR PREVIOUS REPORTS ON JEHMU GREENE:I will NE...  fake
1      WASHINGTON (Reuters) - Congressional Democrats...  real
2      At this point, the arguments against the #AllL...  fake
3      MOSCOW (Reuters) - Russia s Foreign Ministry s...  real
4      One of the largest environmental disasters in ...  fake
...
44893 LONDON (Reuters) - London's status as a global...  real
44894 James Proctor has been identified as the ringl...  fake
44895 WASHINGTON (Reuters) - President Donald Trump ...  real
44896 PARIS (Reuters) - France's far-right National ...  real
44897 This guy just wants everyone dead.Pastor Steve...  fake

```

```
[44898 rows x 3 columns]
```

```
[20]: final_data.head()
```

```

[20]:                                title \
0  FIRE THIS FOX NEWS ANALYST: Trump Was "Alleged...
1  House Democrats escalate effort to obtain Trum...
2  This Tweet Perfectly Connects #AllLivesMatter...
3  Russia's Lavrov and Tillerson talk Syria after...
4  The Worst Environmental Disaster Since The BP...

```

```

                                text      subject \
0  OUR PREVIOUS REPORTS ON JEHMU GREENE:I will NE...  politics
1  WASHINGTON (Reuters) - Congressional Democrats...  politicsNews
2  At this point, the arguments against the #AllL...      News
3  MOSCOW (Reuters) - Russia s Foreign Ministry s...  worldnews
4  One of the largest environmental disasters in ...      News

```

```

                                date label
0      Jan 16, 2017  fake
1  February 14, 2017  real
2      May 21, 2016  fake
3  September 17, 2017  real
4      January 4, 2016  fake

```

```
[21]: final_data.label.value_counts()
```

```

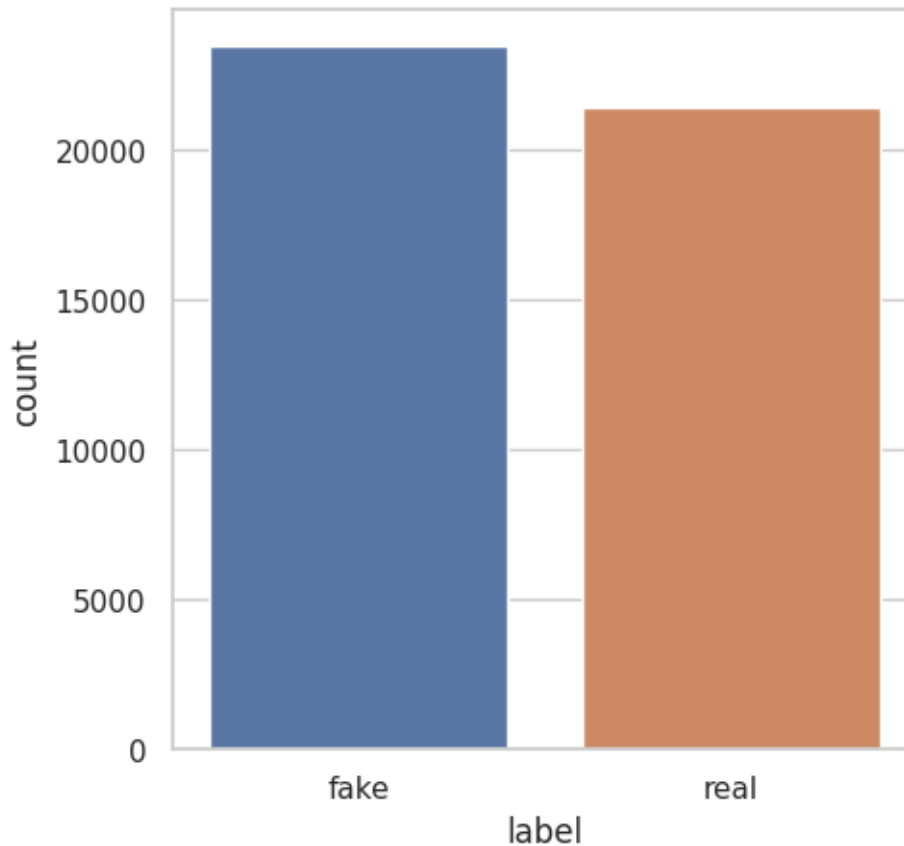
[21]: label
fake      23481
real      21417

```

Name: count, dtype: int64

```
[22]: sns.set_theme(style="whitegrid")
sns.countplot(x=final_data["label"])
```

[22]: <Axes: xlabel='label', ylabel='count'>



```
[23]: final_data.head(5)
```

```
[23]:
```

	title \	text	subject \
0	FIRE THIS FOX NEWS ANALYST: Trump Was "Alleged...		
1	House Democrats escalate effort to obtain Trum...		politics
2	This Tweet Perfectly Connects #AllLivesMatter...		politicsNews
3	Russia's Lavrov and Tillerson talk Syria after...		
4	The Worst Environmental Disaster Since The BP...		News

```

3 MOSCOW (Reuters) - Russia s Foreign Ministry s... worldnews
4 One of the largest environmental disasters in ... News

```

```

          date label
0      Jan 16, 2017  fake
1  February 14, 2017  real
2      May 21, 2016  fake
3  September 17, 2017  real
4      January 4, 2016  fake

```

```
[24]: final_data["text"]=final_data["title"]+final_data["text"]
```

```
[25]: final_data=final_data[["text","label"]]
```

```
[26]: final_data['label'] = final_data['label'].map({'real':1, 'fake':0})
```

```
[27]: final_data.sample(5)
```

```

[27]:
          text  label
43638  MELANIA PUNCHES BACK...HARD...After Leftist Public...      0
11654  Kuwait court sentences MPs to jail terms for s...      1
38270  At G20 summit, Trump pledges $639 million in a...      1
7165   ATLANTA JUDGE Punished for Telling the Truth A...      0
21174  U.S. sanctions North Korean missile experts, R...      1

```

```
[28]: final_data.label[9943]
```

```
[28]: 1
```

```
[29]: final_data.text[36709]
```

```

[29]: ' White Catholic Schoolkids Threaten To Lynch Black Classmate (VIDEO)An as yet
unconfirmed number of white students of a Catholic school in California have
been suspended after sending a video to a black student threatening a
lynching.The Modesto Bee reports:A racist death threat video posted on social
media by a Central Catholic High junior has led to the suspension of students
involved and a criminal investigation, but an African American community group
is calling for calm while police do their work. People are asking me, What can
I do?, but I want to find out what s been done, first. Let the natural process
take its course, said attorney Jacq Wilson, who is working with the family of
the boy targeted in the video.Efforts to dismiss the incident as a high school
prank fell on deaf ears in the community. As Wilson puts it: When has bullying
ever been a joke? When has a death threat ever been a joke? Right now, with
everything going on (nationally), what were they waiting for, for someone to
die? The fact is that people are hurt. People are scared. It s a hate crime. It
s possibly even a domestic act of terrorism. It s a criminal threat, It is also
reassuring that Central Catholic High President Jim Pecchenino came out

```

immediately with his own unambiguous statement on the matter. Students involved in that video have been suspended. They are not on our campus. We ll follow our process for discipline, he said. The whole video is despicable, every part of it. The noose, the gun going off, it s just unconscionable to view. It s very unsettling. The students remain on suspension while the matter is investigated further, with a view to pressing chimerical charges against those responsible. Featured image via Screengrab'

```
[30]: max_len=100
      data_text=final_data["text"]
      data_label=final_data["label"]
```

```
[31]: bert_tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")
```

```
/opt/conda/lib/python3.10/site-
packages/transformers/tokenization_utils_base.py:1601: FutureWarning:
`clean_up_tokenization_spaces` was not set. It will be set to `True` by default.
This behavior will be depracted in transformers v4.45, and will be then set to
`False` by default. For more details check this issue:
https://github.com/huggingface/transformers/issues/31884
  warnings.warn(
```

```
[32]: bert_model = TFBertModel.from_pretrained('bert-base-uncased')
      tokenizer = AutoTokenizer.from_pretrained('bert-base-uncased')
```

Some weights of the PyTorch model were not used when initializing the TF 2.0 model TFBertModel: ['cls.predictions.bias', 'cls.predictions.transform.LayerNorm.bias', 'cls.predictions.transform.dense.bias', 'cls.predictions.transform.LayerNorm.weight', 'cls.seq_relationship.bias', 'cls.seq_relationship.weight', 'cls.predictions.transform.dense.weight'] - This IS expected if you are initializing TFBertModel from a PyTorch model trained on another task or with another architecture (e.g. initializing a TFBertForSequenceClassification model from a BertForPreTraining model). - This IS NOT expected if you are initializing TFBertModel from a PyTorch model that you expect to be exactly identical (e.g. initializing a TFBertForSequenceClassification model from a BertForSequenceClassification model).

All the weights of TFBertModel were initialized from the PyTorch model. If your task is similar to the task the model of the checkpoint was trained on, you can already use TFBertModel for predictions without further training.

```
[33]: X_train, X_test, Y_train, Y_test = train_test_split(data_text, data_label,
      ↪stratify = data_label, test_size = 0.2, random_state =42)
```

```
[34]: def tokenize(X):
```

```

X = bert_tokenizer(
    text = list(X),
    add_special_tokens = True,
    max_length = 100,
    truncation = True,
    padding = 'max_length',
    return_tensors = 'tf',
    return_token_type_ids = False,
    return_attention_mask = True,
    verbose = True
)
return X

```

```

[35]: X_train_token = tokenize(X_train)
      X_test_token = tokenize(X_test)

```

```

[36]: maxlen=100

```

5 create bert model

```

[37]: import tensorflow as tf
      from tensorflow.keras.layers import Input, Dense, Dropout, Layer
      from tensorflow.keras.models import Model

      class BertLayer(Layer):
          def __init__(self, bert_model, **kwargs):
              super(BertLayer, self).__init__(**kwargs)
              self.bert = bert_model

          def call(self, inputs):
              return self.bert(inputs)[1]

      def create_model(bert_model, maxlen=100):
          input_ids = Input(shape=(maxlen,), dtype=tf.int32, name="input_ids")
          attention_mask = Input(shape=(maxlen,), dtype=tf.int32,
                                  name="attention_mask")

          bert_layer = BertLayer(bert_model)
          pooled_output = bert_layer({'input_ids': input_ids, 'attention_mask':
                                  attention_mask})

          x = Dropout(0.5)(pooled_output)
          x = Dense(64, activation="tanh")(x)
          x = Dropout(0.5)(x)
          output = Dense(1, activation="sigmoid")(x)

```

```

    model = Model(inputs=[input_ids, attention_mask], outputs=output)
    return model

# Create the model
model = create_model(bert_model)

# Display the model summary
model.summary()

```

Model: "functional_1"

Layer (type)	Output Shape	Param #	Connected to
attention_mask (InputLayer)	(None, 100)	0	-
input_ids (InputLayer)	(None, 100)	0	-
bert_layer (BertLayer)	(None, 768)	0	attention_mask[0... input_ids[0][0]
dropout (Dropout)	(None, 768)	0	bert_layer[0][0]
dense (Dense)	(None, 64)	49,216	dropout[0][0]
dropout_1 (Dropout)	(None, 64)	0	dense[0][0]
dense_1 (Dense)	(None, 1)	65	dropout_1[0][0]

Total params: 49,281 (192.50 KB)

Trainable params: 49,281 (192.50 KB)

Non-trainable params: 0 (0.00 B)

```

[38]: optimizer = Adam(learning_rate=1e-05, epsilon=1e-08, decay=0.01,clipnorm=1.0)
      model.compile(optimizer = optimizer, loss = 'binary_crossentropy', metrics = _
      ↪ ['accuracy'])

```

/opt/conda/lib/python3.10/site-
packages/keras/src/optimizers/base_optimizer.py:33: UserWarning: Argument

```
`decay` is no longer supported and will be ignored.  
warnings.warn(  

```

```
[39]: from tensorflow.keras.callbacks import EarlyStopping  
  
early_stopping = EarlyStopping(  
    monitor='val_loss', # or another metric you want to monitor  
    patience=3, # number of epochs with no improvement after which training  
    ↪will be stopped  
    restore_best_weights=True  
)  
  
reduce_lr = tf.keras.callbacks.ReduceLROnPlateau(  
    monitor='val_loss',  
    factor=0.1, # Reduce the learning rate by a factor of 0.1  
    patience=3, # Number of epochs with no improvement to wait before reducing  
    ↪the learning rate  
    verbose=1,  
    min_lr=1e-6 # Minimum learning rate  
)
```

```
[40]: history = model.fit(  
    x={  
        'input_ids': X_train_token['input_ids'],  
        'attention_mask': X_train_token['attention_mask']  
    },  
    y=Y_train,  
    epochs=20,  
    batch_size=30,  
    validation_data=(  
        {  
            'input_ids': X_test_token['input_ids'],  
            'attention_mask': X_test_token['attention_mask']  
        },  
        Y_test  
    ),  
    callbacks=[early_stopping, reduce_lr]  
)
```

Epoch 1/20

```
WARNING: All log messages before absl::InitializeLog() is called are written to  
STDERR  
I0000 00:00:1725207084.697593    1327 service.cc:145] XLA service 0x7e2174015270  
initialized for platform CUDA (this does not guarantee that XLA will be used).  
Devices:  
I0000 00:00:1725207084.697645    1327 service.cc:153]   StreamExecutor device  
(0): Tesla P100-PCIE-16GB, Compute Capability 6.0
```

W0000 00:00:1725207085.152191 1327 assert_op.cc:38] Ignoring Assert operator functional_1_1/bert_layer_1/tf_bert_model/bert/embeddings/assert_less/Assert/Assert

2/1198 1:44 87ms/step -
accuracy: 0.5917 - loss: 0.6243

I0000 00:00:1725207091.453288 1327 device_compiler.h:188] Compiled cluster using XLA! This line is logged at most once for the lifetime of the process.

1197/1198 0s 84ms/step -
accuracy: 0.5412 - loss: 0.7820

W0000 00:00:1725207192.856154 1326 assert_op.cc:38] Ignoring Assert operator functional_1_1/bert_layer_1/tf_bert_model/bert/embeddings/assert_less/Assert/Assert

1198/1198 0s 90ms/step -
accuracy: 0.5413 - loss: 0.7819

WARNING: All log messages before absl::InitializeLog() is called are written to STDERR

I0000 00:00:1725207198.797622 1326 asm_compiler.cc:369] ptxas warning :
Registers are spilled to local memory in function 'input_add_reduce_fusion_25',
4 bytes spill stores, 4 bytes spill loads

W0000 00:00:1725207202.064375 1325 assert_op.cc:38] Ignoring Assert operator functional_1_1/bert_layer_1/tf_bert_model/bert/embeddings/assert_less/Assert/Assert

W0000 00:00:1725207228.251592 1324 assert_op.cc:38] Ignoring Assert operator functional_1_1/bert_layer_1/tf_bert_model/bert/embeddings/assert_less/Assert/Assert

1198/1198 157s 116ms/step
- accuracy: 0.5413 - loss: 0.7819 - val_accuracy: 0.7979 - val_loss: 0.5468 -
learning_rate: 1.0000e-05
Epoch 2/20

1/1198 2:08 107ms/step -
accuracy: 0.5000 - loss: 0.7814

I0000 00:00:1725207229.964257 1324 asm_compiler.cc:369] ptxas warning :
Registers are spilled to local memory in function 'input_add_reduce_fusion_24',
4 bytes spill stores, 4 bytes spill loads

1198/1198 125s 105ms/step
- accuracy: 0.6180 - loss: 0.6818 - val_accuracy: 0.8450 - val_loss: 0.4551 -
learning_rate: 1.0000e-05
Epoch 3/20

1198/1198 124s 104ms/step
- accuracy: 0.6821 - loss: 0.6025 - val_accuracy: 0.8677 - val_loss: 0.3913 -
learning_rate: 1.0000e-05

Epoch 4/20
1198/1198 125s 104ms/step
- accuracy: 0.7248 - loss: 0.5452 - val_accuracy: 0.8830 - val_loss: 0.3463 -
learning_rate: 1.0000e-05

Epoch 5/20
1198/1198 125s 104ms/step
- accuracy: 0.7567 - loss: 0.5011 - val_accuracy: 0.8896 - val_loss: 0.3153 -
learning_rate: 1.0000e-05

Epoch 6/20
1198/1198 124s 103ms/step
- accuracy: 0.7764 - loss: 0.4718 - val_accuracy: 0.8983 - val_loss: 0.2911 -
learning_rate: 1.0000e-05

Epoch 7/20
1198/1198 125s 104ms/step
- accuracy: 0.8007 - loss: 0.4387 - val_accuracy: 0.9023 - val_loss: 0.2718 -
learning_rate: 1.0000e-05

Epoch 8/20
1198/1198 125s 104ms/step
- accuracy: 0.8091 - loss: 0.4230 - val_accuracy: 0.9066 - val_loss: 0.2572 -
learning_rate: 1.0000e-05

Epoch 9/20
1198/1198 125s 104ms/step
- accuracy: 0.8218 - loss: 0.4058 - val_accuracy: 0.9104 - val_loss: 0.2451 -
learning_rate: 1.0000e-05

Epoch 10/20
1198/1198 125s 104ms/step
- accuracy: 0.8269 - loss: 0.3937 - val_accuracy: 0.9147 - val_loss: 0.2348 -
learning_rate: 1.0000e-05

Epoch 11/20
1198/1198 125s 104ms/step
- accuracy: 0.8343 - loss: 0.3808 - val_accuracy: 0.9161 - val_loss: 0.2270 -
learning_rate: 1.0000e-05

Epoch 12/20
1198/1198 124s 103ms/step
- accuracy: 0.8381 - loss: 0.3692 - val_accuracy: 0.9186 - val_loss: 0.2190 -
learning_rate: 1.0000e-05

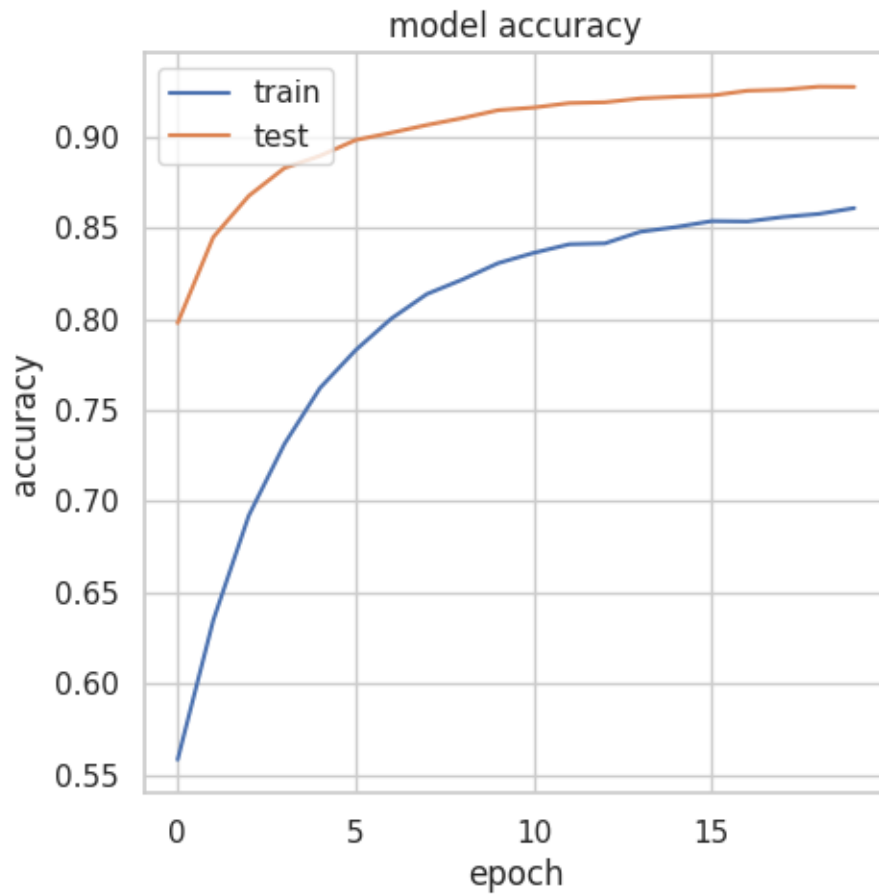
Epoch 13/20
1198/1198 125s 104ms/step
- accuracy: 0.8419 - loss: 0.3600 - val_accuracy: 0.9190 - val_loss: 0.2137 -
learning_rate: 1.0000e-05

Epoch 14/20
1198/1198 125s 104ms/step
- accuracy: 0.8457 - loss: 0.3576 - val_accuracy: 0.9212 - val_loss: 0.2076 -
learning_rate: 1.0000e-05

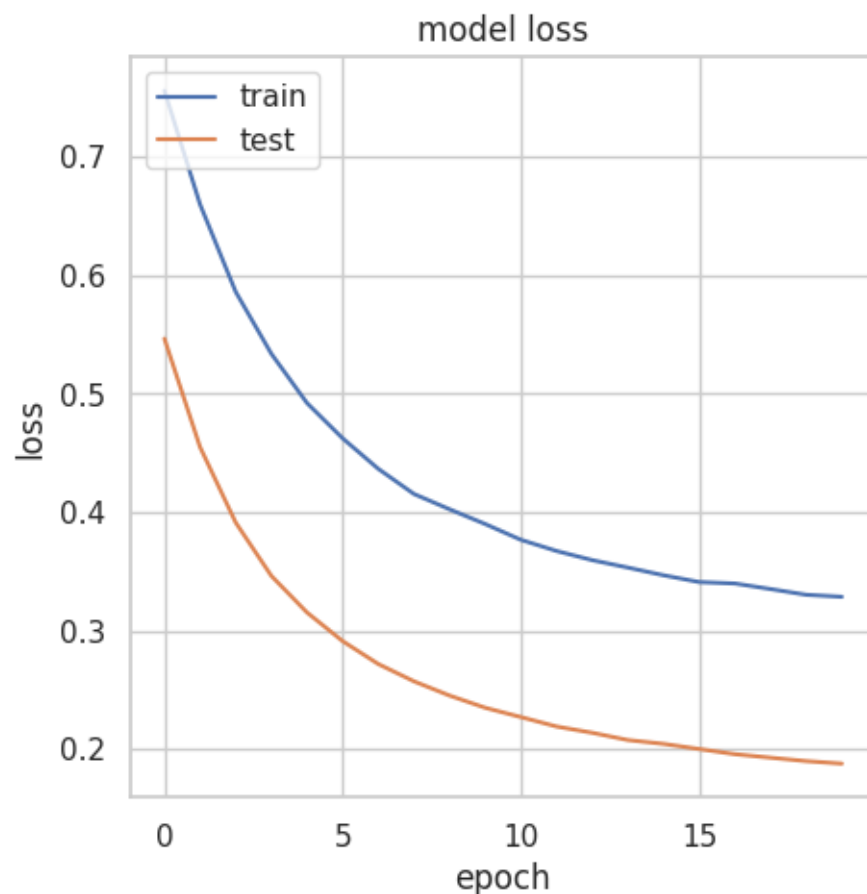
Epoch 15/20
1198/1198 124s 103ms/step
- accuracy: 0.8530 - loss: 0.3441 - val_accuracy: 0.9220 - val_loss: 0.2043 -
learning_rate: 1.0000e-05

```
Epoch 16/20
1198/1198          142s 118ms/step
- accuracy: 0.8508 - loss: 0.3464 - val_accuracy: 0.9227 - val_loss: 0.2000 -
learning_rate: 1.0000e-05
Epoch 17/20
1198/1198          125s 104ms/step
- accuracy: 0.8542 - loss: 0.3413 - val_accuracy: 0.9254 - val_loss: 0.1956 -
learning_rate: 1.0000e-05
Epoch 18/20
1198/1198          124s 103ms/step
- accuracy: 0.8543 - loss: 0.3370 - val_accuracy: 0.9259 - val_loss: 0.1927 -
learning_rate: 1.0000e-05
Epoch 19/20
1198/1198          125s 104ms/step
- accuracy: 0.8577 - loss: 0.3329 - val_accuracy: 0.9276 - val_loss: 0.1898 -
learning_rate: 1.0000e-05
Epoch 20/20
1198/1198          125s 104ms/step
- accuracy: 0.8634 - loss: 0.3255 - val_accuracy: 0.9275 - val_loss: 0.1876 -
learning_rate: 1.0000e-05
```

```
[41]: # summarize history for accuracy
plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])
plt.title('model accuracy')
plt.ylabel('accuracy')
plt.xlabel('epoch')
plt.legend(['train', 'test'], loc='upper left')
plt.show()
```



```
[42]: # summarize history for loss
plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])
plt.title('model loss')
plt.ylabel('loss')
plt.xlabel('epoch')
plt.legend(['train', 'test'], loc='upper left')
plt.show()
```



```
[43]: # Make predictions
y_pred_proba = model.predict({
    'input_ids': X_test_token['input_ids'],
    'attention_mask': X_test_token['attention_mask']
})

# Convert probabilities to binary predictions
y_pred = np.where(y_pred_proba >= 0.5, 1, 0)
```

```
W0000 00:00:1725209619.171169    1327 assert_op.cc:38] Ignoring Assert operator
functional_1_1/bert_layer_1/tf_bert_model/bert/embeddings/assert_less/Assert/Ass
ert
```

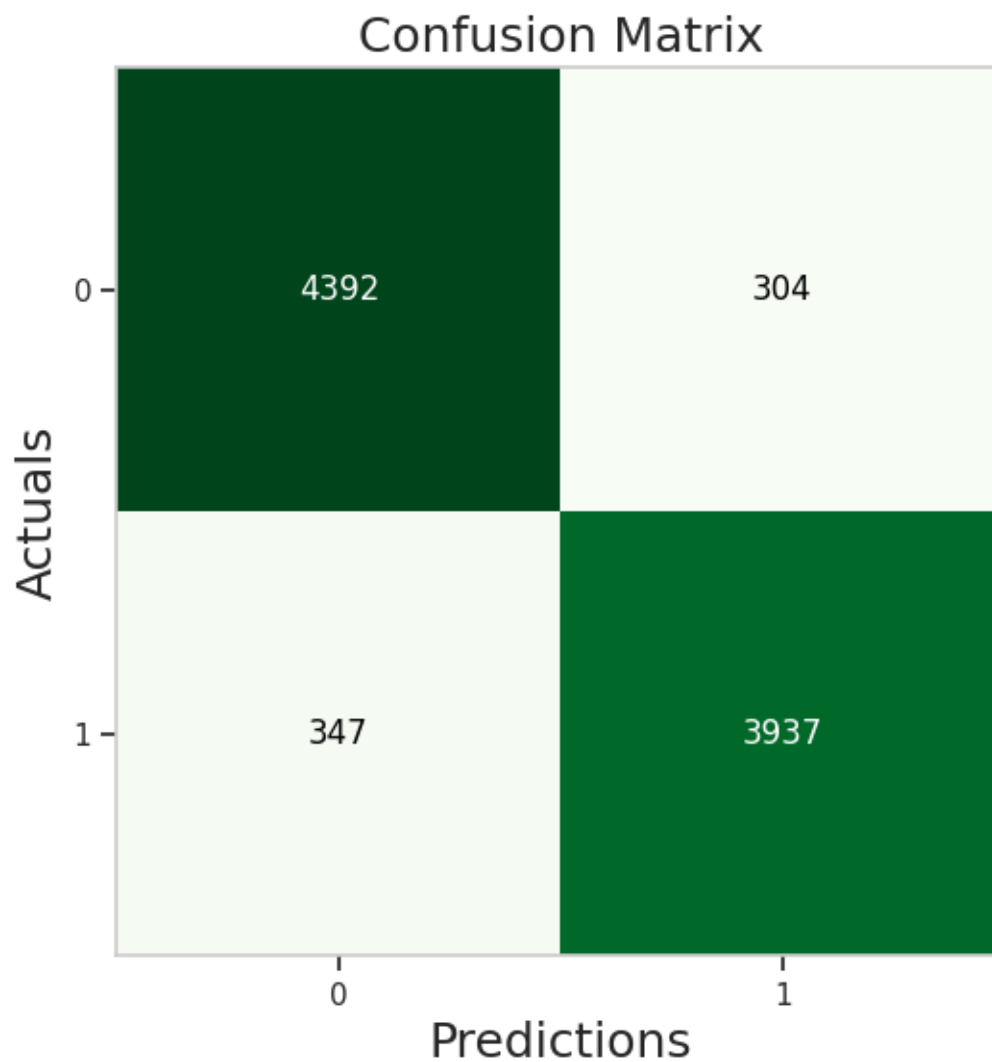
```
280/281          0s 83ms/step
```

```
W0000 00:00:1725209646.696086    1324 assert_op.cc:38] Ignoring Assert operator
functional_1_1/bert_layer_1/tf_bert_model/bert/embeddings/assert_less/Assert/Ass
ert
```

```
281/281          31s 100ms/step
```

6 Confusion matrix

```
[44]: from mlxtend.plotting import plot_confusion_matrix
conf_matrix = confusion_matrix(Y_test,y_pred)
fig, ax = plot_confusion_matrix(conf_mat=conf_matrix, figsize=(6, 6), cmap=plt.
    cm.Greens)
plt.xlabel('Predictions', fontsize=18)
plt.ylabel('Actuals', fontsize=18)
plt.title('Confusion Matrix', fontsize=18)
plt.show()
```



```
[45]: print(classification_report(Y_test,y_pred))
```

precision recall f1-score support

0	0.93	0.94	0.93	4696
1	0.93	0.92	0.92	4284
accuracy			0.93	8980
macro avg	0.93	0.93	0.93	8980
weighted avg	0.93	0.93	0.93	8980

```
[46]: from sklearn.metrics import roc_curve, auc

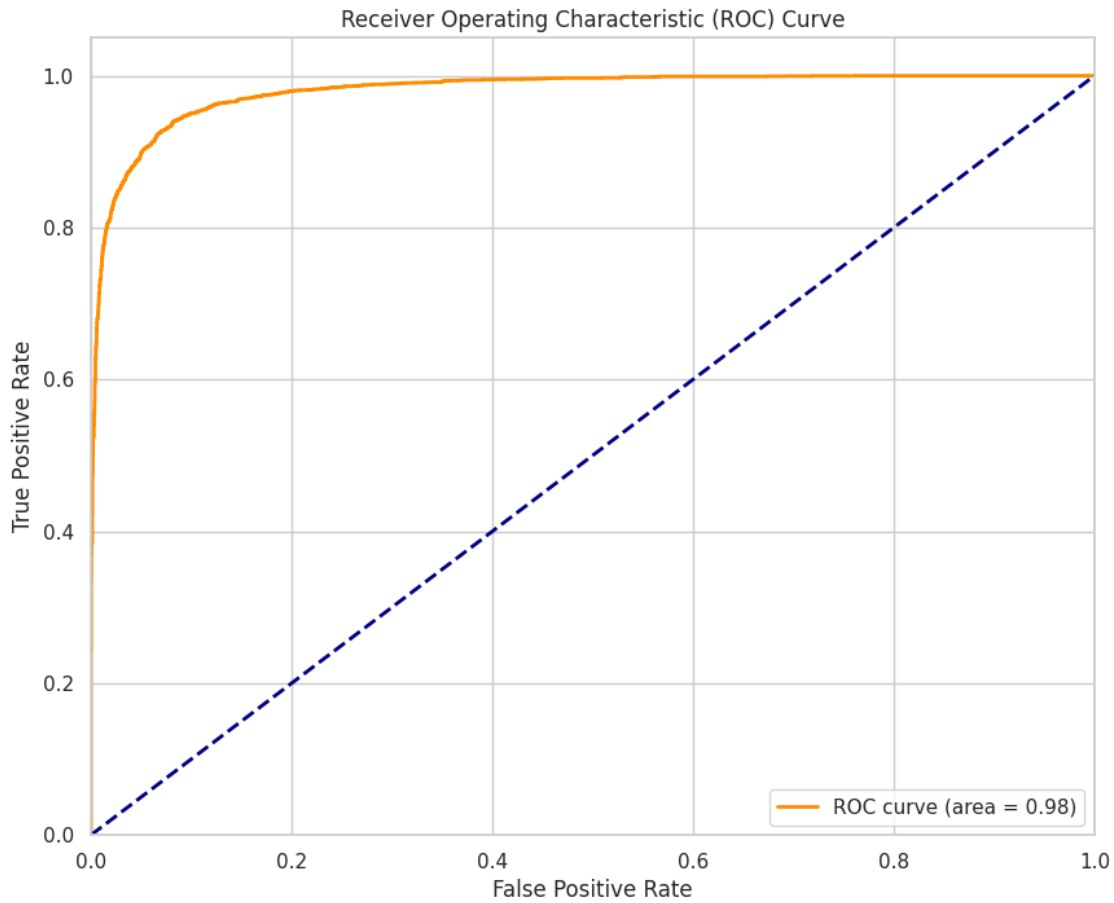
y_pred_prob = model.predict(
    {
        'input_ids': X_test_token['input_ids'],
        'attention_mask': X_test_token['attention_mask']
    }
)

fpr, tpr, thresholds = roc_curve(Y_test, y_pred_prob) # Removed indexing on
↳y_pred_prob
roc_auc = auc(fpr, tpr)

plt.figure(figsize=(10,8))
plt.plot(fpr, tpr, color='darkorange', lw=2, label='ROC curve (area = %0.2f)' %
↳roc_auc)
plt.plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--')
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Receiver Operating Characteristic (ROC) Curve')
plt.legend(loc="lower right")
plt.show()
```

281/281

23s 83ms/step



```
[47]: from sklearn.metrics import precision_recall_curve, auc

y_pred_prob = model.predict(
    {
        'input_ids': X_test_token['input_ids'],
        'attention_mask': X_test_token['attention_mask']
    }
)

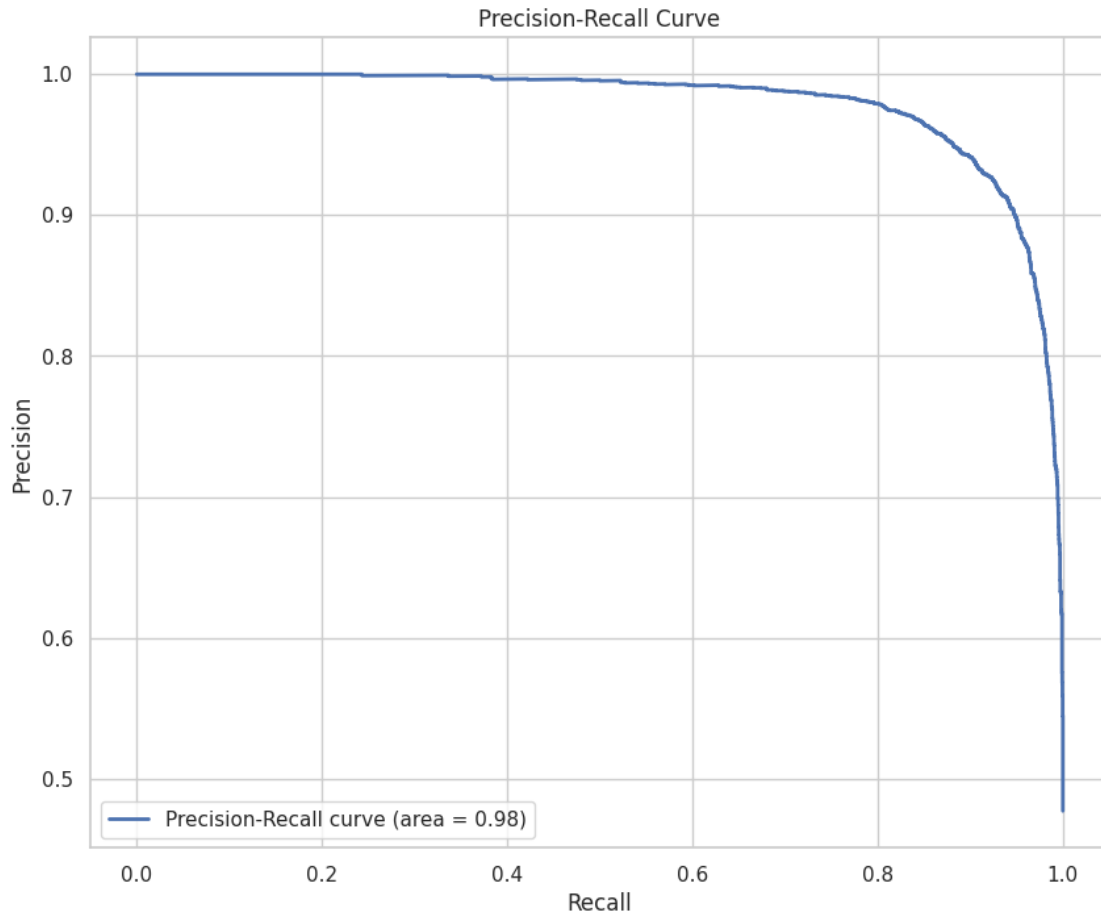
precision, recall, thresholds = precision_recall_curve(Y_test, y_pred_prob)
pr_auc = auc(recall, precision)

plt.figure(figsize=(10,8))
plt.plot(recall, precision, color='b', lw=2, label='Precision-Recall curve_
↪(area = %0.2f)' % pr_auc)
plt.xlabel('Recall')
plt.ylabel('Precision')
```

```
plt.title('Precision-Recall Curve')
plt.legend(loc="lower left")
plt.show()
```

281/281

23s 84ms/step



```
[48]: from sklearn.metrics import accuracy_score

y_pred_prob = model.predict(
    {
        'input_ids': X_test_token['input_ids'],
        'attention_mask': X_test_token['attention_mask']
    }
)
pred = (y_pred_prob > 0.5).astype(int)

accuracy = accuracy_score(Y_test, pred)
```



```
plt.plot([])
plt.text(0, 0, f'Accuracy Score: {accuracy:.4f}', fontsize=16, ha='center',
        va='center', color="indigo")
plt.axis('off')
plt.xlim(-1, 1)
plt.ylim(-1, 1)
plt.show()
```

281/281

23s 84ms/step

Accuracy Score: 0.9275

```
[49]: from sklearn.metrics import roc_auc_score
import matplotlib.pyplot as plt

y_pred_prob = model.predict(
    {
        'input_ids': X_test_token['input_ids'],
        'attention_mask': X_test_token['attention_mask']
    }
)
roc_auc = roc_auc_score(Y_test, y_pred_prob)
```

```
plt.plot([])
plt.text(0, 0, f'ROC AUC Score: {roc_auc:.4f}', fontsize=16, ha='center',
        ↪va='center', color="indigo")
plt.axis('off')
plt.xlim(-1, 1)
plt.ylim(-1, 1)
plt.show()
```

281/281

23s 84ms/step

ROC AUC Score: 0.9800

```
[50]: from sklearn.metrics import cohen_kappa_score, matthews_corrcoef,
        ↪brier_score_loss
import matplotlib.pyplot as plt

# Assuming you have `y_pred_prob` for predicted probabilities and `Y_test` for
        ↪true labels
pred = (y_pred_prob > 0.5).astype(int)

cohen_kappa = cohen_kappa_score(Y_test, pred)
matthews_corr = matthews_corrcoef(Y_test, pred)
brier_score = brier_score_loss(Y_test, y_pred_prob)
```

```

# Plot Cohen's Kappa Score
plt.plot([])
plt.text(0, 0, f'Cohen\'s Kappa Score: {cohen_kappa:.4f}', fontsize=16,
        ↪ha='center', va='center', color="indigo")
plt.axis('off')
plt.xlim(-1, 1)
plt.ylim(-1, 1)
plt.show()

```

Cohen's Kappa Score: 0.8546

```

[51]: # Plot Matthews Correlation Coefficient
plt.plot([])
plt.text(0, 0, f'Matthews Correlation Coefficient: {matthews_corr:.4f}',
        ↪fontsize=16, ha='center', va='center', color="indigo")
plt.axis('off')
plt.xlim(-1, 1)
plt.ylim(-1, 1)
plt.show()

```

Matthews Correlation Coefficient: 0.8547

```
[52]: # Plot Brier Score Loss
plt.plot([])
plt.text(0, 0, f'Brier Score Loss: {brier_score:.4f}', fontsize=16,
        ↪ha='center', va='center', color="indigo")
plt.axis('off')
plt.xlim(-1, 1)
plt.ylim(-1, 1)
plt.show()
```

Brier Score Loss: 0.0541

```
[53]: from sklearn.metrics import log_loss
import matplotlib.pyplot as plt

log_loss_value = log_loss(Y_test, y_pred_prob)

plt.plot([])
plt.text(0, 0, f'Log Loss: {log_loss_value:.4f}', fontsize=16, ha='center',
        ↪va='center', color="indigo")
plt.axis('off')
plt.xlim(-1, 1)
plt.ylim(-1, 1)
plt.show()
```

Log Loss: 0.1876

7 Custom data prediction

[54]:

```
test_text="Cop Shares Racist Meme About Michelle Obama; Now That Cop Is Having
↳A VERY Bad Day (IMAGES)After the election of Donald Trump many folks seem to
↳see it as a permission slip to be as racist and vile as possible. However,
↳here s the thing, you re still going to get called out as racist and vile.
↳And one Alabama police officer just found this out the hard way.According to
↳the Washington Post: Talladega Police Officer Joel Husk was terminated
↳Wednesday for violating the department s social media and code of conduct
↳policies, City Manager Patrick Bryant said. What did he do? So glad you
↳asked: Husk had posted several memes on his Facebook page, including one
↳showing Obama and Melania Trump. Fluent in Slovenian, English, French,
↳Serbian, and German, it said over Trump s photo. Over Obama s, it read:
↳Fluent in Ghetto. Not only that, he posted several extraordinarily racist
↳memes:via Washington Postvia Washington PostAccording to the City Manager,
↳the statements were deemed to be biased or racially insensitive or
↳derogatory and because of that, they have to take action to correct it. If
↳you re going to be a police officer and serve all the public, you can t
↳assume black people standing up for their rights are equivalent to the KKK.
↳That s about the most horrific equivalence imaginable.Also, according to
↳WaPo: Husk, 37, who had been with the department for about two and a half
↳years, had also shared a meme showing President Obama with the words: Was
↳Dallas a terrorist attack? Yes! Carried out by Obama s own homegrown
↳terrorist group! Which is a blatant lie and anyone who were to feel that way
↳belongs nowhere near law enforcement. The city took the proper action
↳letting this racist cop go, and hopefully it will be an example to police
↳departments all over the country that this sort of behavior simply cannot be
↳tolerated.Trump s election must not be allowed to serve as a permission slip
↳to bigots everywhere that it s fine to be as awful as possible, because here
↳in the land of the free and the home of the brave, everyone is protected.
↳Everyone, regardless of color, class, gender, sexual orientation, or creed.
↳Featured Photo by Chip Somodevilla/Getty Images'"
```

```
[55]: test_token = tokenize(test_text)
```

```
[56]: test_text_pred = np.where(model.predict({
    'input_ids' : test_token['input_ids'],
    'attention_mask' : test_token['attention_mask']
})) >= 0.5, 1, 0)
```

68/69

0s 84ms/step

W0000 00:00:1725209751.442950 1326 assert_op.cc:38] Ignoring Assert operator
functional_1_1/bert_layer_1/tf_bert_model/bert/embeddings/assert_less/Assert/Ass
ert

69/69

8s 113ms/step

```
[57]: test_text_pred
```

```
[57]: array([[0],  
            [0],  
            [0],  
            ...,  
            [0],  
            [0],  
            [0]])
```

```
[58]: if(test_text_pred[0]==0):  
        print("News is Fake")  
    else:  
        print("News is Real")
```

News is Fake

```
[ ]:
```